RandomForest To Classify and Predict Non-Diabetic or Prediabetic/Diabetic Individuals Using Lifestyle Information

James Ignacio
Fordham University
155 W 60th Street, New York, NY
973-307-4828
jignacio2@fordham.edu

ABSTRACT

The rising global prevalence of diabetes has underscored the need for accessible and early diagnostic tools. Traditional diabetes prediction models rely heavily on medical metrics such as HbA1c levels, plasma glucose concentration, serum urate, and genetic markers, which are not always easily obtainable for large segments of the population. In this paper, I propose a RandomForest classifier designed to classify individuals as non-diabetic or prediabetic/diabetic based primarily on lifestyle features, without requiring complex medical metrics. Using readily available data from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) such as physical activity, dietary habits, body mass index (BMI), and other lifestyle information, my model aims to offer a more accessible and cost-effective solution for diabetes risk assessment. The model achieved 0.75 accuracy, 0.76 F1 score, and 0.83 AUC, supporting the notion that RandomForest classification is a viable approach for diabetes risk assessment based on lifestyle factors alone. This also shows that the BRFSS is a valid and valuable resource for developing predictive models focused on public health.

1. INTRODUCTION

Diabetes, a chronic disease characterized by elevated blood glucose levels, poses a significant health challenge worldwide. According to the World Health Organization (WHO), diabetes affects an estimated 830 million adults globally, with numbers continuing to rise due to aging populations, increasing obesity rates, and lifestyle changes [9]. In the United States alone, the Centers for Disease Control and Prevention (CDC) reported that as of 2021, 38.4 million Americans were living with diabetes [10]. Alarmingly, 1 in 5 individuals with diabetes and approximately 8 in 10 individuals with prediabetes are unaware of their condition.

Type II diabetes, the most common form of the disease, disproportionately impacts individuals of lower socioeconomic status and its prevalence varies by age, education, income, geographic location, race, and other social determinants of health. The burden of undiagnosed and poorly managed diabetes highlights the urgent need for accessible, scalable, and effective risk assessment tools.

Classification and prediction of diabetes often rely on medical diagnostics, such as glycated hemoglobin (HbA1c), fasting plasma glucose, γ-glutamyltransferase, TCF7L2 and IL6 deleterious alleles, and other biochemical markers [11-12]. While these tests are accurate, they are not universally accessible, particularly in resource-limited settings or for individuals with restricted access to healthcare. This underscores the importance of alternative approaches to assess diabetes risk using more readily available data.

Previous studies have established a connection between diabetes and lifestyle risk factors, including elevated BMI, poor dietary habits, alcohol consumption, smoking, and lack of physical activity [2-5]. The Behavioral Risk Factor Surveillance System (BRFSS) is a health-related telephone survey that is collected annually by the CDC. Each year, the survey collects responses from over 400,000 Americans on health-related risk behaviors, chronic health conditions, and the use of preventative services [13]. It has been conducted every year since 1984. Previous studies have used BRFSS data to predict diseases such as asthma, colorectal cancer, mental distress, and coronary heart disease, as well as find associations between lifestyle behaviors and diseases [14-17].

Machine learning has emerged as a powerful tool in predictive modeling and has been widely explored in the medical field due to its ability to handle large datasets and uncover complex relationships between variables. Models such as RandomForest and CATBoost have demonstrated exceptional performance in predictive tasks, offering robust accuracy, precision, and interpretability. These methods are particularly well-suited for diabetes classification, as they can process numerous variables simultaneously while effectively identifying key predictors [1]. In previous research, RandomForest classifiers have performed well in predicting diabetes [1, 19]. By leveraging lifestyle-related data, such as physical activity, dietary habits, body mass index (BMI), and general health indicators, machine learning offers a promising pathway for developing cost-effective, non-invasive diagnostic tools.

This project aims to classify individuals as non-diabetic or prediabetic/diabetic using a RandomForest classifier trained on lifestyle-related features from the BRFSS, showing that lifestyle features are an effective predictor for diabetes.

2. Materials and Methods

2.1. Dataset

The dataset utilized in this study originates from the Behavioral Risk Factor Surveillance System (BRFSS), a telephone survey conducted annually by the Centers for Disease Control and Prevention (CDC). The BRFSS collects responses from a representative sample of over 400,000 Americans, capturing information on health-related risk behaviors, chronic health conditions, and the use of preventative services. This dataset provides a rich source of lifestyle and health-related information, making it suitable for developing machine learning models aimed at classifying diabetes risk.

For this project, the 2015 BRFSS dataset available on Kaggle was used, containing 70,692 entries with 21 features and 2 target classes: non-diabetic and prediabetic/diabetic. The dataset is balanced, ensuring that both target classes are equally represented. The dataset includes a diverse range of features related to health, lifestyle, and demographic factors. These features are encoded numerically, making the dataset inherently compatible with scikit-learn, which will be used for the RandomForest classifier. Many of these features are binary, yes or no, answers to questions asked on the survey. A full description of the data can be found on Kaggle [18].

2.2. RandomForest Classification

RandomForest is a popular ensemble learning method primarily used for classification and regression tasks in machine learning. Developed by Leo Breiman in 2001, it builds upon the concept of decision trees by introducing randomness and aggregation to improve performance and reduce overfitting. It is both robust and versatile, making it suitable for a wide range of applications, including diabetes risk assessment [6-7].

RandomForest is composed of multiple decision trees, each functioning as an individual predictive model that makes decisions by recursively partitioning the data based on feature values. While single decision trees are susceptible to overfitting, RandomForest mitigates this issue by combining the predictions of numerous trees. As an ensemble method, it aggregates the outputs of these individual models to generate a final prediction. Each tree is trained on a unique random subset of the data, typically created through bootstrapping (sampling with replacement), which enhances model diversity and reduces overfitting.

2.3 Performance Measurement

The performance of the RandomForest model is evaluated using a range of metrics to ensure a comprehensive assessment of its predictive capabilities.

Accuracy is one of the most commonly used evaluation metrics in machine learning due to its simplicity and ease of interpretation. Accuracy refers to the proportion of correctly predicted samples in the total sample. Generally, the higher the accuracy, the better the model's performance. The formula for calculating accuracy is as follows:

$$Accuracy = (TP + TN)/(TP + FP + TN + FN)$$

where *TN* stands for True Negative, *FN* for False Negative, *TP* for True Positive, and *FP* for False Positive.

Precision is a metric that measures how many of the predicted positive instances are actually true positives. In other words, it tells us the accuracy of positive predictions made by the model. The formula for precision is:

$$Precision = TP/(TP + FP)$$

Recall (Sensitivity) represents the proportion of actual positive samples correctly identified by the model. A higher sensitivity indicates a model's ability to correctly identify positive instances, reducing the likelihood of missed diagnoses. The formula for Recall is:

$$Recall = TP/(TP + FN)$$

F1 Score is the harmonic mean of precision and sensitivity. It is useful in situations where there is an imbalance between precision and recall. A higher F1 score indicates a balance between precision and recall. The formula for F1 score is:

$$F1 = (2 \times (Precision \times Recall))/(Precision + Recall)$$

The ROC Curve (Receiver Operating Characteristic curve) is a graphical representation that helps evaluate the performance of a binary classification model across various thresholds. It plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at different classification thresholds.

The area under the curve (AUC) provides a single value that summarizes the model's ability to discriminate between positive and negative cases.

$$TPR = (TP + FN)/TP$$

$$FPR = FP/(FP + TN)$$

To optimize the model's performance, Randomized Search Cross-Validation (RandomizedSearchCV) is employed for hyperparameter tuning. This involves randomly sampling a specified number of hyperparameter combinations within defined ranges and evaluating each combination using cross-validation. This approach efficiently identifies the optimal hyperparameters while reducing computational costs compared to exhaustive grid search methods [8]. After this, an exhaustive grid search is performed using Grid Search Cross-Validation (GridSearchCV) on a more narrow set of parameters depending on the RandomizedSearchCV.

3. Results

3.1 Best Model Parameters

Randomized Search Cross-Validation (RandomizedSearchCV) and GridSearch Cross-Validation (CV) were employed for hyperparameter tuning. The parameter grid includes the following as described in the table below:

Table 2. Parameter grid description for RandomizedSearchCV and GridSearchCV

Parameter	Description
n_estimators	Number of trees in the forest
max_depth	Maximum depth of each tree
min_samples_split	Minimum samples required to split an internal node
min_samples_leaf	Minimum samples required to be at a leaf node
criterion	Function to measure the quality of a split (gini or entropy)
max_features	Number of features to consider for the best split (sqrt or log2)

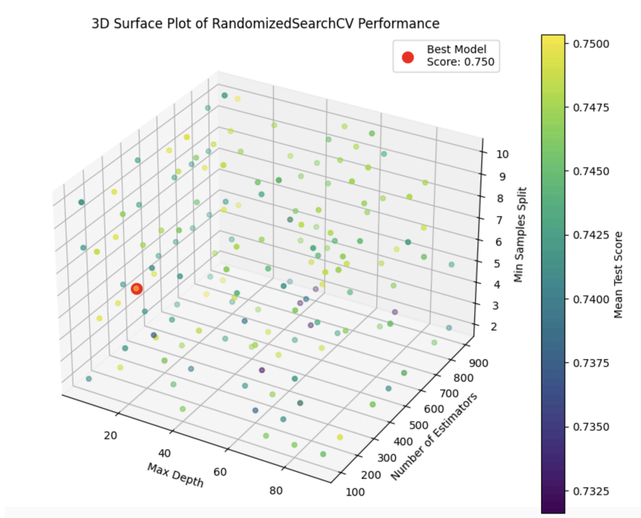
bootstrap	Whether bootstrap samples are used
class_weight	Whether to address any imbalance or not

The first RandomizdSearchCV consisted of sampling 200 random combinations from the parameter grid. Each combination is used to create a model, which is then evaluated using 5-fold cross-validation. The performance of each model is measured using the F1 score, and the model with the best F1 score is selected. F1 score was chosen over accuracy score despite the dataset being balanced because the cost of false positives and false negatives is not equal. A false negative, predicting that a person does not have diabetes when they actually do, has more implications than a false positive, predicting that a person has diabetes when they do not.

To visualize the impact of different hyperparameter configurations on model performance, we created a 3D scatter plot, where each point represents a unique combination of hyperparameters (max_depth, n_estimators, and min_samples_split) and its corresponding mean test score.

RandomizedSearchCV stores the results in the cv_results_ attribute, which is a dictionary. This can be used to create a DataFrame that can be visualized. The color of each point reflects the mean test score, with a gradient scale to emphasize performance differences. The best-performing model, identified by the highest mean test score, is highlighted in red, providing a clear visual reference for the optimal parameter set. This model achieved the highest performance with a mean test score of 0.75 across the 5-fold cross validation.

Figure 1. 3D Surface Plot indicating model performance using different parameters in the RandomizedSearchCV



The GridSearchCV then focuses on a more narrow set of parameters based on what the RandomizedSearchCV found was optimal. The GridSearchCV searches every possible combination of the parameters instead of randomly selecting 200. This approach is particularly effective because RandomizedSearchCV reduces the initial computational cost by sampling a subset of the parameter grid. Then, GridSearchCV hones in on the most relevant combinations, leveraging the insights gained from the random search phase. This step-by-step refinement ensures a balance between thoroughness and computational efficiency.

The search found that the best model consisted of the following parameters as shown in Table 2:

Table 3. Optimal Parameters for RandomForest via RandomizedSearchCV and GridSearchCV

Parameter	Value
n_estimators	300
max_depth	10
min_samples_split	5
min_samples_leaf	1
criterion	gini
max_features	sqrt
bootstrap	True
class_weight	balanced

3.2 Model Performance

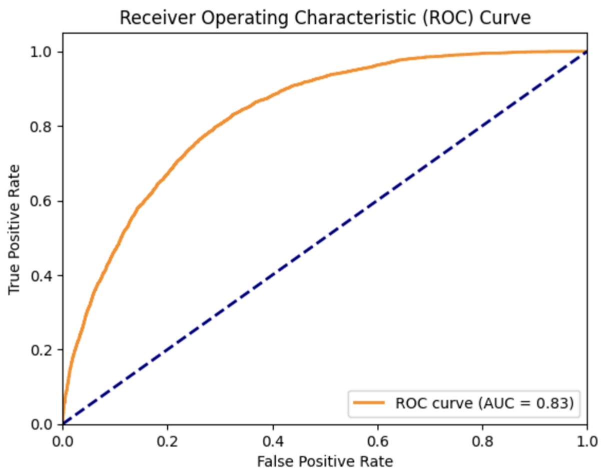
The model achieved an accuracy of 0.75, meaning it correctly classified 75% of the instances in the test set. This indicates a fair level of performance. The model achieved a precision of 0.73 and a recall of 0.80. Precision indicates that among the predicted positive cases, 73% were actually positive. This suggests that the model has a good ability to correctly identify positive cases but may have some issues with false positives. On the other hand, recall indicates that among the actual positive cases, 80% were correctly identified by the model. This indicates that the model is effective at capturing most of the actual positive cases but may miss some of them (false negatives). The F1 score for the model is 0.76, which balances the precision and recall, suggesting the model provides a good balance between the ability to capture positive cases and to minimize false positives and false negatives. The metrics are organized in the table below:

Table 4. Results of the RandomForest Classification

Metric	Value
accuracy	0.75
F1 score	0.76
precision	0.73
recall	0.80

The model also achieved an AUC of 0.83. This indicates that the model has strong discriminative power in distinguishing between positive and negative cases. The Figure below depicts the ROC curve, with the AUC value of the model shown.

Figure 2. Area Under the Receiver Operating Characteristic Curve of RandomForest Classifier



The model performed better than a baseline model that did not undergo RandomSearchCV and GridSearchCV. The baseline model achieved the following results:

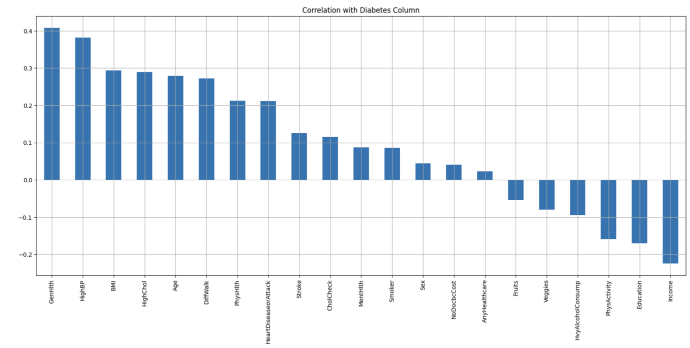
Table 5. Results of the Baseline RandomForest Classification

Metric	Value
accuracy	0.73
F1 score	0.74
precision	0.72
recall	0.78
AUC	0.81

3.3 Important Features

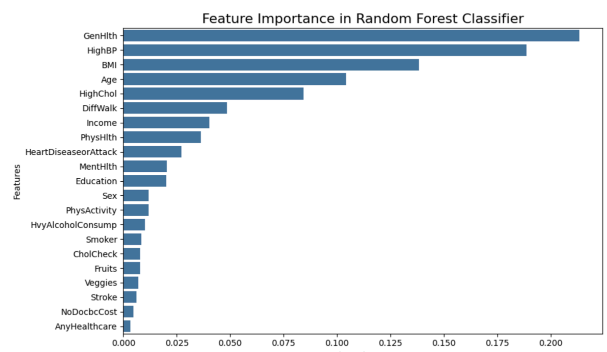
In the exploratory analysis, a bar chart was constructed using the Pearson correlation coefficient metric to assess correlations between features in the dataset. This analysis identified HighBP (High Blood Pressure), GenHlth (General Health), HighChol (High Cholesterol), BMI (Body Mass Index), Age, and DiffWalk (Difficulty Walking) were some features positively correlated with the Diabetes feature. Income, Education, and PhysActivity (Physical Activity) were some features negatively correlated with the Diabetes feature. Below is the described chart:

Figure 3. Chart of the correlation of dataset features to diabetes



Additionally, feature importance scores were obtained using the feature_importances_ attribute from the scikit-learn RandomForest Classifier. feature_importances_ is calculated using Gini importance, which quantifies the contribution of each feature to impurity reduction at decision nodes across all trees in the forest. The importance of each feature is determined by the total decrease in node impurity, weighted by the number of samples reaching that node. These feature importances are then averaged across all trees to provide a mean value, with standard deviation reflecting the variability in their impact [20]. Below a graph indicating the features that were most important in classification:

Figure 4. Bar chart of important features in RandomForest Classification



GenHlth, HighBP, BMI, Age, and HighChol were the five most important features in classification.

4. Discussion

The RandomForest Classifier achieved great results, with an accuracy of 0.75, indicating that it can make sound predictions on the test set. It also achieved an F1 score of 0.76, suggesting that the model provides a reasonable balance between finding true positives and avoiding false positives. It achieved a precision of 0.73 and a recall of 0.80, indicating that the model does a good job of capturing most individuals with diabetes, minimizing the risk of false negatives with a recall of 0.80, but also showing that the model's ability to reduce false positives is not as well-performing with a precision of 0.73.

The Area Under the Receiver Operating Characteristic Curve (AUC) of 0.83 further supports the model's discriminative power, demonstrating its strong ability to distinguish between positive and negative cases. This performance metric is particularly valuable in clinical settings where accurate prediction of disease status is critical.

In the exploratory analysis, the bar chart constructed using the Pearson correlation coefficient identified key features such as HighBP (High Blood Pressure), GenHlth (General Health), HighChol (High Cholesterol), BMI (Body Mass Index), Age, and DiffWalk (Difficulty Walking) as most positively correlated with the Diabetes feature. The feature_importances_ attribute indicates GenHlth, HighBP, BMI, Age, and HighChol as the five most influential in classification. This supports their relevance as predictors of diabetes, underscoring their contribution to the model's predictive capability.

The identification of these features as the most important predictors of diabetes is a significant validation of the model's performance. These features align closely with established medical and epidemiological knowledge, where these factors are well-documented contributors to the risk of diabetes. High Blood Pressure and High Cholesterol are both components of metabolic syndrome, a cluster of conditions that significantly increases the risk of type 2 diabetes [21-22]. BMI is a direct measure of body fat and obesity, which are among the most prominent predictors of diabetes, particularly type 2 [21]. Age is universally recognized as a key factor, with the likelihood of diabetes increasing with age due to cumulative risk exposure and metabolic changes [22]. Difficulty Walking reflects mobility and lifestyle limitations, which are often associated with poorer physical activity and higher risks for metabolic conditions [21]. General Health is a measure of overall health, which can indirectly reflect underlying conditions such as hypertension, hyperlipidemia, or obesity, all of which contribute to the development of type 2 diabetes. By identifying these features as key contributors, the model demonstrates that it is leveraging medically and biologically relevant information, enhancing its interpretability and trustworthiness.

These results suggest that the RandomForest model, with its balanced accuracy, AUC, and feature importance, provides a robust framework for diabetes prediction. The model's performance, combined with the identification of key features, can guide clinical decision-making by prioritizing interventions and resource allocation based on an individual's risk profile.

5. Conclusions

The global rise in diabetes underscores the need for accessible and cost-effective diagnostic tools. This study demonstrated the potential of a RandomForest classifier to classify individuals as non-diabetic or prediabetic/diabetic using lifestyle data from the 2015 BRFSS. The model achieved strong performance metrics, including an accuracy of 0.75, an F1 score of 0.76, and an AUC of 0.83, highlighting its effectiveness in diabetes risk assessment without requiring complex medical metrics. Key predictors identified include General Health, High Blood Pressure, BMI, Age, and High Cholesterol, emphasizing the role of lifestyle factors. These results suggest that machine learning models using accessible data from sources like the BRFSS can support early diabetes detection and contribute to public health efforts.

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Data Availability Statement: All data in this paper can be downloaded at https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset/data. All code can be accessed at https://github.com/jamesignac/diabetes_rfc_pred/tree/main.

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